
Baby Ginger

Baby ginger is one of the most lucrative crops per unit area. It is highly dependent on heat and that is why we grow it in our heated greenhouse. Ginger takes a long time to grow, but it is well worth it as it is a low maintenance and high yielding crop. The rarity of baby ginger provides a lot of excitement to customers.

Family

- Zingiberaceae

Cultivars

- Use mature organic ginger as your seeds. Those can be bought in any food store. Any cultivar works.

Crop Planning

We plan for one generation of ginger per season. The crop will take a month and a half in the nursery and 4 months in the ground to be ready for harvest. In our climate, the peak growing season of baby ginger is from April to September.

Ginger can be transplanted into an unheated tunnel or greenhouse, but best yields are achieved in a heated greenhouse (in temperate climates).

Starting Seeds

- 88 lbs (40 kg) of ginger seeds is needed for one bed. Those need to be started as seedlings in the nursery before being transplanted to get a jump start on the long season required to grow ginger.
- Buy seeds 6 weeks ahead of the transplant date. If seeds are bought earlier than that, keep them in a cool room at 54 °F (12 °C).

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- From seed to transplant, plan for a 4 to 6 week period depending on light availability and capacity to maintain heat for seedlings. Seeds are best started in March for a transplant date in April and a first harvest in August.
 - When ready to seed, cut ginger seeds in pieces of 100g each. Avoid seeding smaller pieces. Make cuts on the narrower section of ginger.
 - Fill 4 in (10 cm) pots with a mix of 70% high fertility potting mix and 30% compost. Leave 1 in (2,5 cm) unfilled. Deposit 1 seed per pot. Cover the seed with a small layer of potting mix and water the seedlings.
 - Keep seedlings moist, but do not over water, since it could induce rotting of the seed.
 - Germinate seeds at a temperature of 75-82 °F (24 to 28 °C). Use heating mats in order to get to that heat. In addition, a clear plastic or a row cover placed on seedlings can also help germination. On sunny days, take off the clear plastic or row cover, if the temperature under it goes over 104 to 113 °F (40 to 45 °C). Take off completely the clear plastic or row cover once seeds are germinated.

Intensive Spacing

1 row per bed, spaced every 4 in (10 cm) on the row.

Fertilization

- To grow well, ginger needs high doses of nutrients (especially nitrogen and potassium). Under fertilizing will lead to poor rhizome development.
- Three fertilizations are needed. One at transplant and one at each of the two hillings. Use a high nitrogen fertilizer like pelleted chicken manure (5-3-2) and add one dose of the tomato fertilizing mix (compost (2-1-1), alfalfa meal (2-0-2), feather meal (11-0-0) Sul-Po-Mag (0-0-22) and potassium sulfate (0-0-50).

Irrigation

Irrigation Type

- Drip tape (2 lines per bed), placed on each side of the row.

Irrigation Management

- Root depth: Shallow (15-24 in [38-61 cm])
- Likes steady supply of water. Can be on the same irrigation schedule as tomatoes, eggplants or peppers. Avoid over-watering ginger.
- General guideline : on a hot and sunny summer day, give 6 water cycles of 10 minutes each. Soil should be moist, but never soaked.

Implantation

Ginger is ready to be transplanted in the greenhouse as soon as the soil is at 50 °F (10 °C) or more and the seedlings are approximately 11 in (30 cm) high. It should be around mid-April.

Equipment

- Broadfork
- Fertilization
- Power harrow
- 30 in (75 cm) bed prep rake and tubes
- Shovel
- Irrigation equipment (drip tape)

Itinerary

- Each bed is prepared just before planting as it makes transplanting easier (since the soil is very loose after being harrowed).
- Remove the tarp and use the rake to quickly clean each bed if there is a lot of large debris.
- Use a broadfork on each bed.

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- Apply fertilization on the bed and pass with the power harrow to a depth of 2 in (5 cm).
 - Dig an 8 in (20 cm) trench on the center of the bed. Leave a lot of loose soil around the trench since ginger needs hilling.
 - Using a hose, soak the trench with water and bring the well-irrigated ginger pots along each bed.
 - Transplant the 4 in (10 cm) pots in the trench all stucked together (4 in [10 cm] spacing). Make sure to cover all transplants with 1 in (2,5 cm) of soil, but also keep some loose soil for later hillings.
 - Install 2 drip tapes per bed. One per side and start the irrigation for one hour after the transplant, then keep the soil moist for the first week after planting.

Climate Management

- Ideal temperature range for ginger growth is 60-80 °F (15-26 °C).
- If problematic to achieve, or in order to cut heating cost, a second structure of clear plastic can be installed above the growing plants. To do this, install a clear plastic with holes on two galvanized steel wires hung high along the bed. Place a heating tube under this additional layer.
- It is also a good strategy to place the ginger in the center of the greenhouse, where the climate is hotter and more humid.

Weeding and Maintenance

- To avoid weeding, soil around the ginger plants can be recovered with a white plastic.
- Since ginger is a very competitive plant, two hand-weeding should be enough to control weeds (if your soil is not covered).
- Ginger should be hilled and fertilized twice in the season. Each time, the idea is to add at least 4 inches of soil on top of the ginger, mixing in the amendments.

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- On the second pass, the plants are already quite big, and the hilling can be done with a rotary plow to speed up the process.

Plant Protection

- Before transplant, a preventive application on seedlings of *Trichoderma harzianum* (Rootshield) is recommended to avoid fungal diseases like Pythium. Apply the recommended dosage by the manufacturer. If leaves are overly yellow or wilting, it could be caused by those diseases.
- After transplant, there are very few pests that affect ginger. Look out for spider mites and fungus gnats, although the risk is fairly low.

Harvest

- Start harvesting your crops four months after the initial transplant in order to sell baby ginger (pink skin with juicy and tender flesh). At that stage, the ginger will have developed big rhizomes, but won't have yet developed its rough skin (each seed generating a rhizome at least 4 times its size). If rhizomes are too small, a few growing weeks can be added before harvest.
- Harvest ginger by first loosening the rhizomes in the soil with a fork. Then, slowly pull the ginger out of the soil by pulling on the leaves.
- Since ginger can stay in the soil for many weeks without damage, harvest each bed progressively. Harvest each week the quantity that you are able to sell until there is none left. Note that after 10 months in the soil, ginger is at full maturity. Between month 4 and month 10 it will grow its beige rough skin.

Conditioning

- Rinse rhizomes.
- Cut roots with scissors or cutters.
- Cut rhizomes to the size needed and rinse again until no more soil is left.
- Leaves can be sold with the rhizomes or cut off.

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- 1 person should package 44 lbs (20 kg) of ginger per hour.

Storing

- Harvest the day of your market/delivery or a day before. Store ginger inside a plastic bag placed in a closed container at a temperature around 36 °F (2 °C).